45)

$r_n = \frac{(0.529)n^2}{z}$ Å $\quad E_n = \frac{-13.6\,(z^2)}{n^2}$ ev atom^{-1}

for Li^{2+} z = 3

Bohr raduis for the third orbit (r_3)

$= \frac{(0.529)\,(3)^2}{3}$

$= 0.529 \times 3$

$= 1.587$ Å

Energy of an electron in the fourth orbit (E_4) $= \frac{-13.6\,(3)^2}{(4)^2}$

$= -7.65$ eV atom^{-1}

46) Given

$v = 2.85 \times 10^8$ ms^{-1}

$m_p = 1.673 \times 10^{-27}$ Kg

$\lambda = \frac{h}{mv}$

$= \frac{6.626\times10^{-34}\,\text{kgm}^2\text{s}^{-1}}{1.673\times10^{-27}\,\text{kg}\times2.85\times10^{8}\,\text{ms}^{-1}}$

$\lambda = 1.389\times10^{-15} \Rightarrow \lambda = 1.389\times10^{-5}$ Å

[$\because 1$ Å $= 10^{-10}$ m]

47) m = 160 g = 160×10^{-3} kg

v = 140 km hr^{-1} = $\frac{140\times10^3}{60\times60}$ ms^{-1}

v = 38.88 ms^{-1}

$\lambda = \frac{h}{mv}$

$\lambda = \frac{6.626\times10^{-34}\,\text{Kgm}^2\text{s}^{-1}}{160\times10^{-3}\,\text{Kg}\times38.88\,\text{ms}^{-1}}$

$\lambda = 1.065\times10^{-34}$ m

48) $\Delta x = 0.6$ Å $= 0.6 \times 10^{-10}$ m

Δp = ?

$\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$

$\Delta x \,.\, \Delta p \geq 5.28\times10^{-35}\,\text{kgm}^2\text{s}^{-1}$

$(0.6\times10^{-10})\Delta p \geq 5.28\times10^{-35}$

$\Rightarrow \Delta p \geq \frac{5.28\times10^{-35}\,\text{kgm}^2\text{s}^{-1}}{0.6\times10^{-10}\,\text{m}}$

$\Delta p \geq 8.8\times10^{-25}$ kgms^{-1}

49)

$\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$

$\lambda.(m\Delta v) \geq \frac{h}{4\pi}$

$\Delta v \geq \frac{h}{4\pi(m\lambda)}$

$\Delta v \geq \frac{h}{4\pi\times m\times\frac{h}{mv}}$ $\quad \left[\because \lambda = \frac{h}{mv}\right]$

$\Delta v \geq \frac{v}{4\pi}$

Therefore,

minimum uncertainity in velocity $= \frac{v}{4\pi}$

50) Potential difference = 100V

$= 100 \times 1.6 \times 10^{-19}$J

$\lambda = \frac{h}{\sqrt{2mev}}$

$= \frac{6.626\times10^{-34}\,\text{Kgm}^2\text{s}^{-1}}{\sqrt{2\times9.1\times10^{-31}\,\text{Kg}\times100\times1.6\times10^{-19}\,\text{J}}}$

$\lambda = 1.22\times10^{-10}$ m

51)

n	l	m_l	sub energy level
4	**2**	0	4d
3	1	0	**3p**
5	**1**	**any one value −1, 0, +1**	5p
3	**2**	−2	3d

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